ENTERPRISE KNOWLEDGE INTELLIGENCE PLATFORM

Team integration workflow, shared contracts, responsibilities and 24-hour execution plan

6. HOW ALL FIVE MODULES CONNECT

- 1. Person 5's frontend sends a document upload request.
- 2. Person 1's FastAPI backend receives the request.
- 3. Person 2's ingestion pipeline parses the document and creates structured chunks.
- 4. Person 3 indexes the chunks using embeddings and keyword search.
- 5. Person 5's frontend submits an employee question.
- 6. Person 1's LangGraph orchestrator analyzes the question.
- 7. Person 1 selects local Ollama or the permitted cloud model.
- 8. The orchestrator invokes Person 4's MCP tools when needed.
- 9. Person 4's MCP server calls the appropriate document and retrieval services.
- 10. Person 3 returns relevant evidence.
- 11. Person 1 generates an answer using the retrieved evidence.
- 12. The answer is checked for citation support and coverage.
- 13. Person 5 displays the final response and citations.

7. SHARED API AND DATA CONTRACTS

All members must use a common data format. Every retrieved passage must preserve:

- chunk_id
- document_id
- document_name
- version
- is_current
- section
- page
- content
- access_scope

Example retrieved passage

```
{
  "chunk_id": "chunk_001",
  "document_id": "policy_001",
  "document_name": "Travel Policy",
  "version": "2.0",
  "is_current": true,
  "section": "5.2",
  "page": 12,
  "content": "Supporting policy passage..."
}
```

The final chat response should contain:

```
{
  "answer": "The manager can approve expenses up to the specified limit...",
  "status": "partially_answered",
  "citations": [
    {
      "document_id": "policy_001",
```

```
      "document_name": "Travel Policy",
      "version": "2.0",
      "section": "5.2",
      "page": 12,
      "chunk_id": "chunk_001"
    }
  ],
  "verification": {
    "verified": true,
    "unsupported_claims": []
  }
}
```

These are proposed interface contracts. The actual answer and status must always reflect the retrieved evidence.

8. INTEGRATION RESPONSIBILITIES

Integration	Responsible members
Backend + document ingestion	Person 1 + Person 2
Ingestion + database/indexing	Person 2 + Person 3
Retrieval + MCP	Person 3 + Person 4
MCP + LangGraph	Person 4 + Person 1
Ollama + LangGraph	Person 1
Verification + answer pipeline	Person 1 + Person 5
Frontend + FastAPI	Person 5 + Person 1
Complete end-to-end integration	All five

Person 1 is the overall technical integration lead. Each member remains responsible for integrating their own module with the adjacent module.

9. 24-HOUR DEVELOPMENT SCHEDULE

Time	Main objective
0–2 hours	Setup, repository, architecture and contracts
2–6 hours	Independent module development
6–9 hours	First integrations
9–12 hours	Core RAG assistant working
12–15 hours	MCP, Ollama and LangGraph integration
15–18 hours	Version intelligence, GraphRAG and verification
18–21 hours	Frontend integration and full testing
21–24 hours	Feature freeze, bug fixes and final demo

Mandatory milestones

- Hour 6: Every member has an independently running module.
- Hour 9: Ingestion and retrieval are connected.
- Hour 12: Core document-question-answering system works with citations.
- Hour 15: MCP and hybrid AI are integrated.
- Hour 18: All major components are connected.

- Hour 21: Feature freeze.
- Hour 24: Final demonstration.

10. TEAM DEVELOPMENT RULES

- Every member owns their assigned module.
- No one should independently change shared schemas without agreement.
- Each person works on a separate Git branch.
- Person 1 manages the main integration branch.
- Use mock data while waiting for dependent modules.
- Test modules independently before integration.
- Integrate every 2–3 hours.
- Share working progress with the team regularly.
- Do not spend hours adding optional technologies before core functionality works.
- Reserve the final three hours for stability and demonstration.

FINAL TEAM OWNERSHIP

Person	Primary ownership
Person 1	AI Brain, Backend, Ollama, LangGraph and Overall Integration
Person 2	Document Intelligence and Ingestion
Person 3	Retrieval, Database, Embeddings and GraphRAG
Person 4	MCP Server and AI Tool Integration
Person 5	Frontend, User Experience and Product Testing

Our objective is to develop one integrated, reliable Enterprise Knowledge Intelligence Platform, not five disconnected modules.

Each member must deliver a working component that can be independently demonstrated and successfully integrated into the complete application.